

PAPA, Hungary

WMO: 128250

Lat: 47.364N

Lon: 17.501E

Elev: 499

StdP: 14.43

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.0	(c) 17.6	(d) 6.4	(e) 7.8	(f) 15.5	(g) 11.7	(h) 10.2	(i) 20.5	(j) 26.5	(k) 34.5	(l) 23.2	(m) 36.3	(n) 5.5	(o) 350	(p) 0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	90.7	70.0	87.5	69.2	84.2	68.0	72.2	85.6	70.6	83.6	69.2	81.6	7.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.9	104.5	78.1	66.3	98.5	76.1	64.7	93.2	74.5	36.1	85.8	34.8	84.1	33.6	81.6	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.9	(b) 19.6	(c) 16.9	(e) 7.0	(f) 94.4	(g) 6.3	(h) 3.9	(i) 2.5	(j) 97.2	(k) -1.2	(l) 99.5	(m) -4.7	(n) 101.7	(o) -9.3	(p) 104.5		
(a) 22.9	(b) 19.6	(c) 16.9	(e) 5.7	(f) 75.2	(g) 6.6	(h) 1.5	(i) 1.0	(j) 76.3	(k) -2.8	(l) 77.1	(m) -6.5	(n) 77.9	(o) -11.3	(p) 79.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	32.1	35.5	42.9	52.6	60.0	67.0	70.2	69.3	61.0	51.9	43.5	34.5	
	DBStd	15.17	8.14	8.71	8.00	7.27	6.47	6.39	5.87	5.87	6.35	7.52	8.12	7.51	
	HDD50	2047	556	409	244	56	4	0	0	0	2	68	225	483	
	HDD65	5332	1020	826	686	374	182	52	20	23	151	407	645	946	
	CDD50	2701	1	3	23	134	313	510	627	598	332	128	30	2	
	CDD65	513	0	0	0	3	26	112	182	156	31	3	0	0	
	CDH74	5356	0	0	0	45	305	1164	1895	1627	305	15	0	0	
	CDH80	1823	0	0	0	3	47	371	716	627	59	0	0	0	
Wind	WSAvg	6.9	6.9	7.8	8.0	8.0	7.2	6.5	6.4	5.5	6.0	6.1	6.9	7.2	
Precipitation	PrecAvg	26.5	1.4	1.2	1.5	1.9	2.4	3.0	2.8	2.8	2.4	1.9	2.1	2.1	
	PrecMax	39.1	2.8	2.7	3.4	4.0	7.8	6.3	5.9	8.3	6.2	4.3	3.8	3.8	
	PrecMin	17.6	0.2	0.1	0.3	0.2	0.7	0.3	0.4	0.1	0.7	0.0	0.3	0.2	
	PrecStd	5.3	0.7	0.8	0.9	1.0	1.5	1.5	1.3	1.9	1.7	1.2	0.8	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	60.6	68.4	79.0	84.3	91.6	94.8	95.8	85.7	76.1	67.0	58.6	
		MCWB	48.4	49.3	53.0	60.9	66.1	70.6	71.6	69.8	66.6	63.2	57.4	51.9	
	2%	DB	50.9	55.6	64.3	74.5	80.8	87.9	90.5	90.8	81.1	71.6	62.6	52.1	
		MCWB	45.3	47.3	51.5	58.0	64.2	70.0	70.0	69.8	65.5	60.8	54.9	47.1	
	5%	DB	46.7	51.9	60.3	70.4	77.3	84.3	87.4	86.9	77.4	68.0	58.9	47.9	
		MCWB	42.5	45.0	49.9	56.4	62.9	68.9	69.2	68.9	64.0	59.0	53.0	44.2	
	10%	DB	43.4	48.2	55.9	66.7	73.7	80.8	84.0	82.8	73.6	64.2	55.2	44.6	
		MCWB	40.0	43.0	47.5	54.4	61.0	66.8	68.0	67.4	62.4	57.2	50.4	41.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	50.2	54.9	62.1	68.4	73.7	74.3	73.8	68.8	64.2	58.0	51.9
			MCDB	54.8	58.8	65.3	76.4	80.7	87.1	88.2	87.0	81.1	73.3	65.6	57.5
2%		WB	45.9	47.8	52.5	59.5	66.1	71.4	72.2	71.7	67.0	61.8	55.5	47.5	
		MCDB	50.5	54.7	62.0	72.0	77.8	84.0	85.7	85.6	77.9	69.7	61.7	52.0	
5%		WB	42.8	45.6	50.4	57.3	63.9	69.7	70.6	70.1	65.2	59.9	53.2	44.4	
		MCDB	46.1	50.9	58.9	68.3	75.0	81.6	83.4	83.6	74.9	66.6	58.3	47.1	
10%		WB	40.3	43.3	48.2	55.3	62.0	67.9	69.1	68.5	63.3	57.8	50.8	41.9	
		MCDB	43.1	47.5	55.4	65.1	72.0	78.9	81.5	80.5	71.7	63.4	54.6	44.0	
Mean Daily Temperature Range	5% DB	MDBR	10.7	13.7	17.2	19.8	19.7	20.5	21.4	21.6	19.1	17.1	12.2	10.0	
		MCDBR	15.2	20.2	23.9	26.1	25.4	25.2	26.7	27.8	25.0	21.4	16.7	14.6	
		MCWBR	11.0	13.0	14.0	13.6	12.5	10.8	10.0	10.6	11.6	12.3	10.8	10.9	
	5% WB	MCDBR	14.8	17.9	21.8	23.9	22.8	23.3	23.8	24.6	21.9	19.8	16.3	13.3	
		MCWBR	10.9	12.1	13.2	13.0	12.0	10.7	10.1	10.5	10.8	11.7	10.8	10.3	
Clear-Sky Solar Irradiance	taub	0.327	0.353	0.390	0.436	0.429	0.431	0.439	0.441	0.417	0.386	0.358	0.328		
	taud	2.363	2.319	2.237	2.127	2.201	2.234	2.213	2.219	2.253	2.317	2.346	2.378		
	Ebn at Noon	235	253	260	259	266	265	261	255	248	237	220	220		
	Edh at Noon	23	30	37	46	44	43	43	41	36	30	23	20		
All-Sky Solar Radiation	RadAvg	347	604	988	1440	1719	1869	1832	1626	1160	738	391	266		
	RadStd	35	84	119	139	168	129	141	160	135	73	39	33		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.86	N/A	N/A	N/A	N/A	-173	N/A	+142	+88
(47) Regional (74 neighbors)		+1.08	N/A	N/A	+1.64	+0.94	+0.83	-220	-311	+184

Nomenclature: See separate page